

	resultant ()	[1] answer
4(a)(i)	$P = IV$ $36 = I(12)$ $I = 3.0 \text{ A}$	[1] for equation and substitution [1] for correct answer
4(a)(ii)	$Q = It$ $= (3.0)(20 \times 60)$ $= 3600 \text{ C}$	[1] for correct equation and substitution [1] for correct answer

No.	Solution	Remarks
4(a)(iii)	$E = Pt$ $= (36)(20 \times 60)$ $= 43200 \text{ J}$	[1] for equation, substitution and correct answer
4(a)(iv)	$R = \frac{V}{I}$ $= \frac{12}{3.0}$ $= 4.0 \, \Omega$	[1] for equation, substitution and correct answer
4(b)(i)	$\frac{1}{R_{\text{eff}}} = \frac{1}{4.0} \times 3$ $R_{\text{eff}} = 1.33 \, \Omega$ $I = \frac{12}{1.33} = 9.0 \text{ A}$	[1] for correct R_{eff} [1] for correct current
4(b)(ii)1.	There is no change to the potential difference across the lamps since they are all connected in parallel to an ideal battery. With the same potential difference and resistance, the power dissipated and brightness remain unchanged.	[1] correct explanation [1] correct conclusion
4(b)(ii)2.	Due to the internal resistance of the battery, the <u>potential difference across the lamps decreases</u> . Hence, the <u>brightness of the lamps decreases</u> .	[1] p.d. across lamps decreases [1] brightness decreases
5(a)	(400)	